CI/CD Pipeline & Monitoring Stack

A real-world DevOps project built on live VPS infrastructure

2	VPS Servers
5	Project Phases
7	Alert Rules

Docker | GitHub Actions | Prometheus | Grafana | Nginx | Alertmanager

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Linux System Administrator | RHCSA Certified

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About Me

KETUL PATEL

🎓 Certification

RHCSA — Red Hat Certified System Administrator
RHCE Training Completed
AWS Solutions Architect Training
Oracle Cloud Foundations Associate

💼 Experience

Managing production Linux & Windows servers
Server deployment, configuration & troubleshooting
System security hardening & access control
Backup validation & recovery (R1Soft)

⚙️ Core Skills

Linux
RHEL, Ubuntu, AlmaLinux, CentOS

Web Servers
Apache, Nginx, IIS

Cloud
AWS EC2, S3, IAM, VPC

Monitoring
Prometheus, Grafana, Zabbix

CI/CD
GitHub Actions, Docker

Networking
DNS, TCP/IP, Firewall, SSH

Project Overview

What was built — end to end

01
App & Docker

Flask app containerised

✅ Done

02
GitHub Actions

Auto CI/CD on git push

✅ Done

03
Nginx Proxy

Port 80 with SSL ready

✅ Done

04
Prometheus & Grafana

Linux + Windows monitoring

✅ Done

05
Alertmanager

Slack alerts + Email escalation

✅ Done

Infrastructure:

🐧 Linux VPS — Ubuntu 24.04 LTS — App server, CI/CD target, Prometheus + Grafana hub

🪟 Windows VPS — Windows Server 2025 — Monitoring target, windows_exporter agent

Architecture

GitHub Repository

Source code + Actions workflows

GitHub Actions (CI/CD)

lint → test → build → SSH deploy

SSH deploy

🐧 Linux VPS — Ubuntu 24.04

- Nginx (port 80 → 5000)
- Docker → Flask App
- Prometheus (metrics hub)
- Grafana (dashboards)
- Alertmanager (Slack + Email)

scrapes metrics

🪟 Windows VPS — WS 2025

- windows_exporter (port 9182)
- IIS / test application
- Self-hosted runner (optional)

Phase 1 & 2 — App, Docker & CI/CD

01 Flask App + Docker

- Python Flask app with /health & /metrics endpoints
- Dockerfile — python:3.12-slim base image
- prometheus-flask-exporter for real metrics
- docker build -t devops-app .
- docker run -d -p 5000:5000 devops-app

✅ curl localhost:5000 → {"status":"running","version":"1.0"}

✅ curl localhost:5000/health → {"status":"healthy"}

02 GitHub Actions CI/CD

- Triggers on every git push to main branch
- Runs automated tests first
- Builds new Docker image
- SSHes into Linux VPS via secrets
- Stops old container, deploys new one
- Zero manual intervention needed

✅ git push → Actions → Docker build → VPS deploy

✅ Version 1.0 → 2.0 → 3.0 auto deployed

Phase 3 & 4 — Nginx & Monitoring

03 Nginx Reverse Proxy

- Nginx installed on Linux VPS
- Configured as reverse proxy on port 80
- Forwards traffic to Docker app on port 5000
- proxy_set_header for real client IPs
- App accessible via public VPS IP
- SSL/Certbot ready for domain setup

✅ http://Linux_VPS _IP → Flask app responds

04 Prometheus + Grafana

Prometheus

Metrics collection hub — Linux + Windows

node_exporter

Linux VPS — CPU, RAM, disk, network

windows_exporter

Windows VPS — full server metrics

Grafana

Dashboard ID 1860 (Linux) + custom (Windows)

Flask metrics

/metrics endpoint via prometheus-flask-exporter

✅ Both VPS servers live in Grafana dashboards

Phase 5 — Alertmanager

Slack + Email escalation

Alert Fires
up == 0 detected by Prometheus

→

Slack #1
Immediate notification

→

Slack ×5
Repeats every 5 minutes

→

Email
Escalation after 25 minutes

→

Resolved
Auto-clears when server recovers

Alert Rules Configured

InstanceDown	Any server unreachable > 1 min
HighCPULinux	Linux CPU > 85% for 5 min
HighCPUWindows	Windows CPU > 85% for 5 min
HighMemoryLinux	Linux RAM > 90% for 5 min
HighMemoryWindows	Windows RAM > 90% for 5 min
LowDiskLinux	Linux disk < 10% free
LowDiskWindows	Windows disk < 10% free

Results & What Was Achieved

5	2	7	3
Phases Completed	VPS Servers Monitored	Alert Rules Active	App Versions Auto Deployed

What This Project Proves

- ✓ Can build a real CI/CD pipeline from scratch — not just follow a tutorial
- ✓ Understands Docker containerisation and deployment automation
- ✓ Can monitor multi-OS infrastructure (Linux + Windows) with open-source tools
- ✓ Configured production-grade alerting with escalation logic
- ✓ Documents work professionally — GitHub repo + live documentation site
- ✓ Uses real VPS servers — not just a local VM or sandbox environment

Tech Stack

Application
- Python Flask
- Docker
- prometheus-flask-exporter

CI/CD
- GitHub Actions
- SSH Deploy
- Git

Web Server
- Nginx
- Reverse Proxy
- Port 80/443

Monitoring
- Prometheus
- Grafana
- node_exporter

Windows Mon.
- windows_exporter
- Custom Dashboard

Alerting
- Alertmanager
- Slack Webhook
- Gmail SMTP

Infrastructure
- Ubuntu 24.04 LTS
- Windows Server 2025
- 2× VPS

Version Control
- GitHub
- GitHub Pages
- Jekyll

Let's Connect

Open to DevOps, Linux SysAdmin & Infrastructure roles

💼 LinkedIn	linkedin.com/in/ketulrpatel
🐙 GitHub	github.com/ketulpa
🌐 Portfolio	ketulpa.github.io
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🏆 RHCSA Certified | RHCE Training | AWS Solutions Architect Training | Oracle Cloud Foundations Associate